

VISUALIZATION LIBRARY DOCUMENTATION

Matplotlib and Seaborn

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INTRODUCTION

Data visualization is the process of representing data in a graphical format such as charts, graphs, and plots. It helps users understand patterns, trends, and relationships within data more effectively than raw numerical values.

Python offers several powerful visualization libraries including Matplotlib, Seaborn, Plotly, Bokeh, and Pandas. These libraries provide different tools and functionalities for creating meaningful visual representations of data.

This documentation focuses on Matplotlib and Seaborn, two of the most widely used visualization libraries in Python. It discusses their features, graph types, practical examples, use cases, and comparison.

MATPLOTLIB

Matplotlib is a popular open-source data visualization library in Python. It provides extensive capabilities for creating static, animated, and interactive visualizations. It is widely used in scientific computing, data analysis, and business reporting.

Key Features

- Supports multiple chart types
- Highly customizable plotting options
- Easy integration with NumPy and Pandas
- Suitable for publication-quality visualizations
- Strong community support and documentation

Common Use Cases

- Trend analysis
- Business reporting
- Scientific research
- Data exploration
- Dashboard creation

LINE PLOT

Description

A line plot is used to display information as a series of data points connected by straight lines. It is useful for showing trends and changes over a period of time.

Use Case

Tracking monthly sales, stock market trends, temperature changes, and website traffic.

Example

```
import matplotlib.pyplot as plt
```

```
months = ["Jan", "Feb", "Mar", "Apr", "May"]
```

```
sales = [120, 150, 180, 170, 210]
```

```
plt.plot(months, sales, marker='o')
```

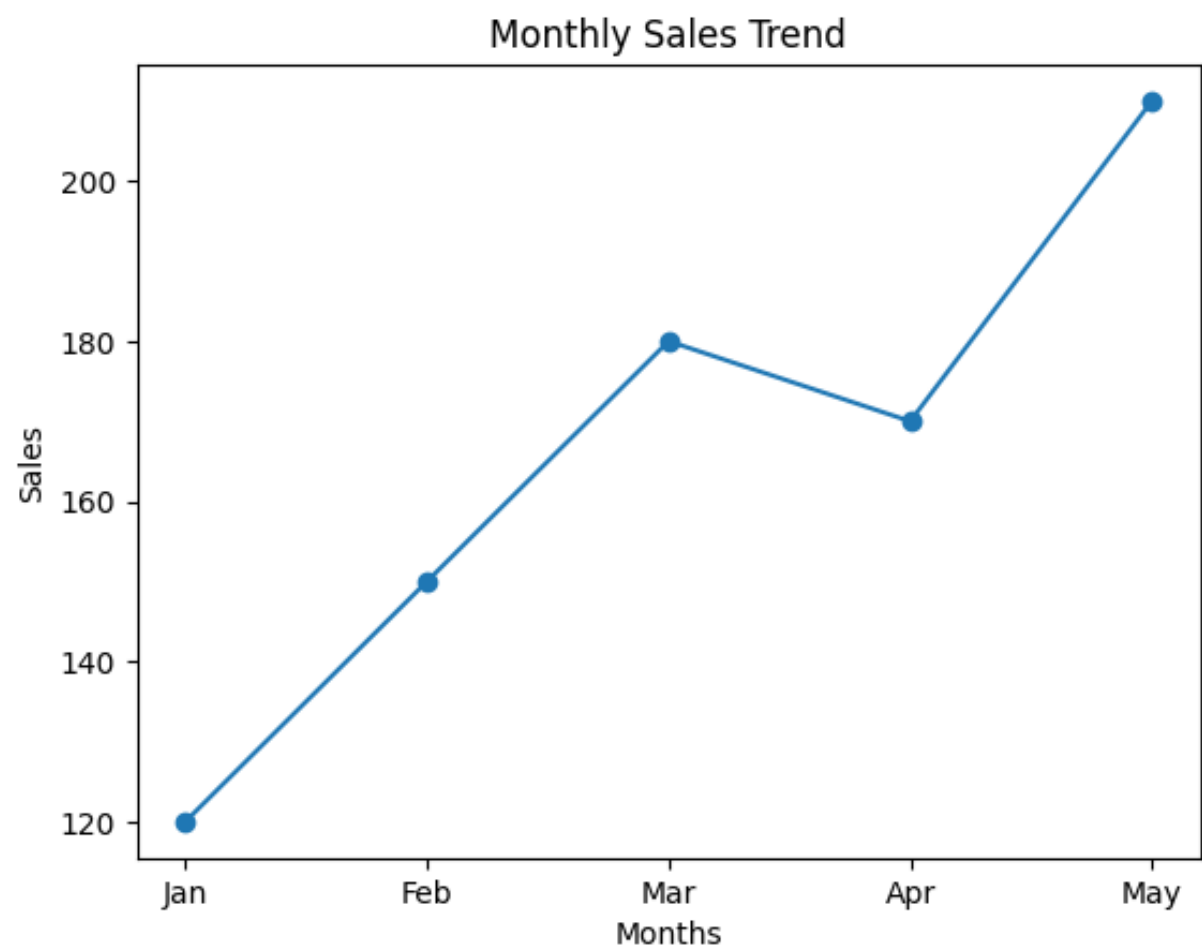
```
plt.title("Monthly Sales Trend")
```

```
plt.xlabel("Months")
```

```
plt.ylabel("Sales")
```

```
plt.show()
```

Output



SCATTER PLOT

Description

A scatter plot displays individual data points on a graph to show the relationship between two numerical variables.

Use Case

Analyzing relationships between study hours and exam scores, advertising cost and revenue, or height and weight.

Example

```
import matplotlib.pyplot as plt
```

```
hours = [1, 2, 3, 4, 5, 6]
```

```
scores = [45, 50, 60, 70, 80, 90]
```

```
plt.scatter(hours, scores)
```

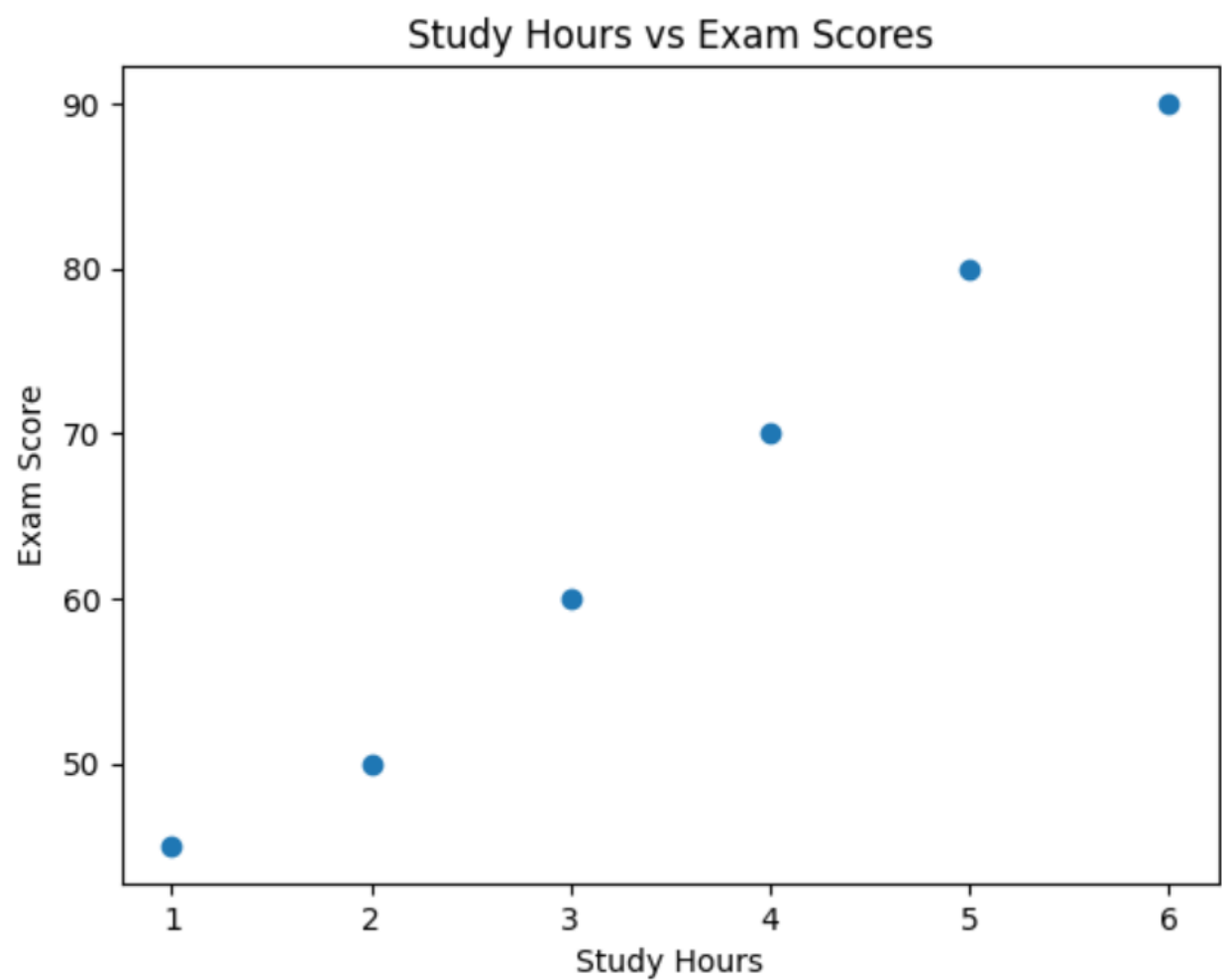
```
plt.title("Study Hours vs Exam Scores")
```

```
plt.xlabel("Study Hours")
```

```
plt.ylabel("Exam Score")
```

```
plt.show()
```

Output



PIE CHART

Description

A pie chart represents data as slices of a circle, where each slice shows the proportion of a category relative to the whole dataset.

Use Case

Displaying market share, budget allocation, survey results, or product sales percentages.

Example

```
import matplotlib.pyplot as plt
```

```
labels = ["Product A", "Product B", "Product C", "Product D"]
```

```
sales = [35, 25, 20, 20]
```

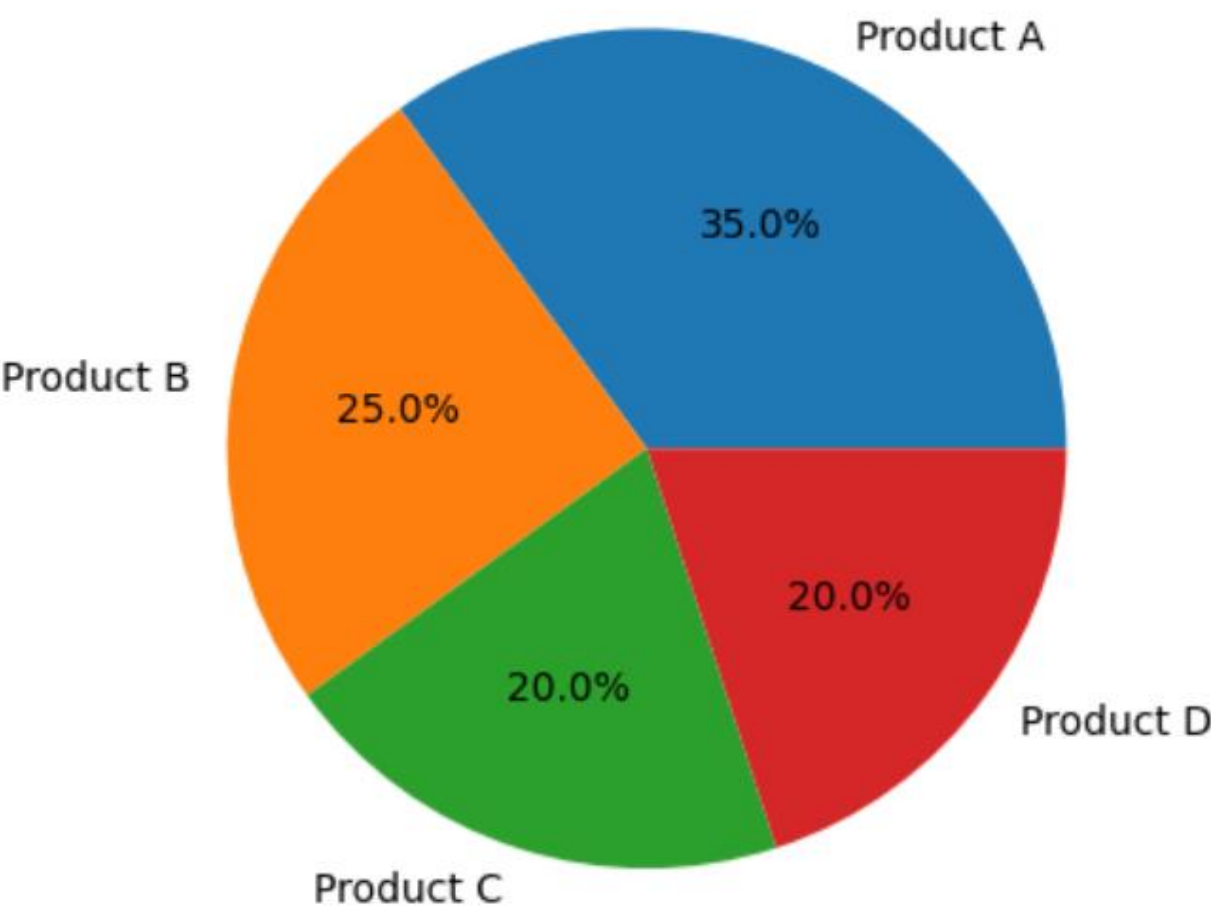
```
plt.pie(sales, labels=labels, autopct='%1.1f%%')
```

```
plt.title("Product Sales Distribution")
```

```
plt.show()
```

Output

Product Sales Distribution



SEABORN

Seaborn is a high-level Python visualization library built on top of Matplotlib. It simplifies the process of creating attractive and informative statistical graphics through built-in themes and advanced plotting functions.

Key Features

- Attractive default themes and color palettes
- Easy statistical visualization
- Built-in support for complex plots
- Seamless integration with Pandas DataFrames
- Simplified plotting syntax

Common Use Cases

- Exploratory Data Analysis
- Statistical reporting
- Correlation analysis
- Distribution analysis
- Data reporting and dashboards

BAR PLOT

Description

A bar plot represents categorical data using rectangular bars. It is commonly used to compare values across different categories.

Use Case

Comparing sales across regions, department performance, or customer preferences.

Example

```
import seaborn as sns
```

```
import matplotlib.pyplot as plt
```

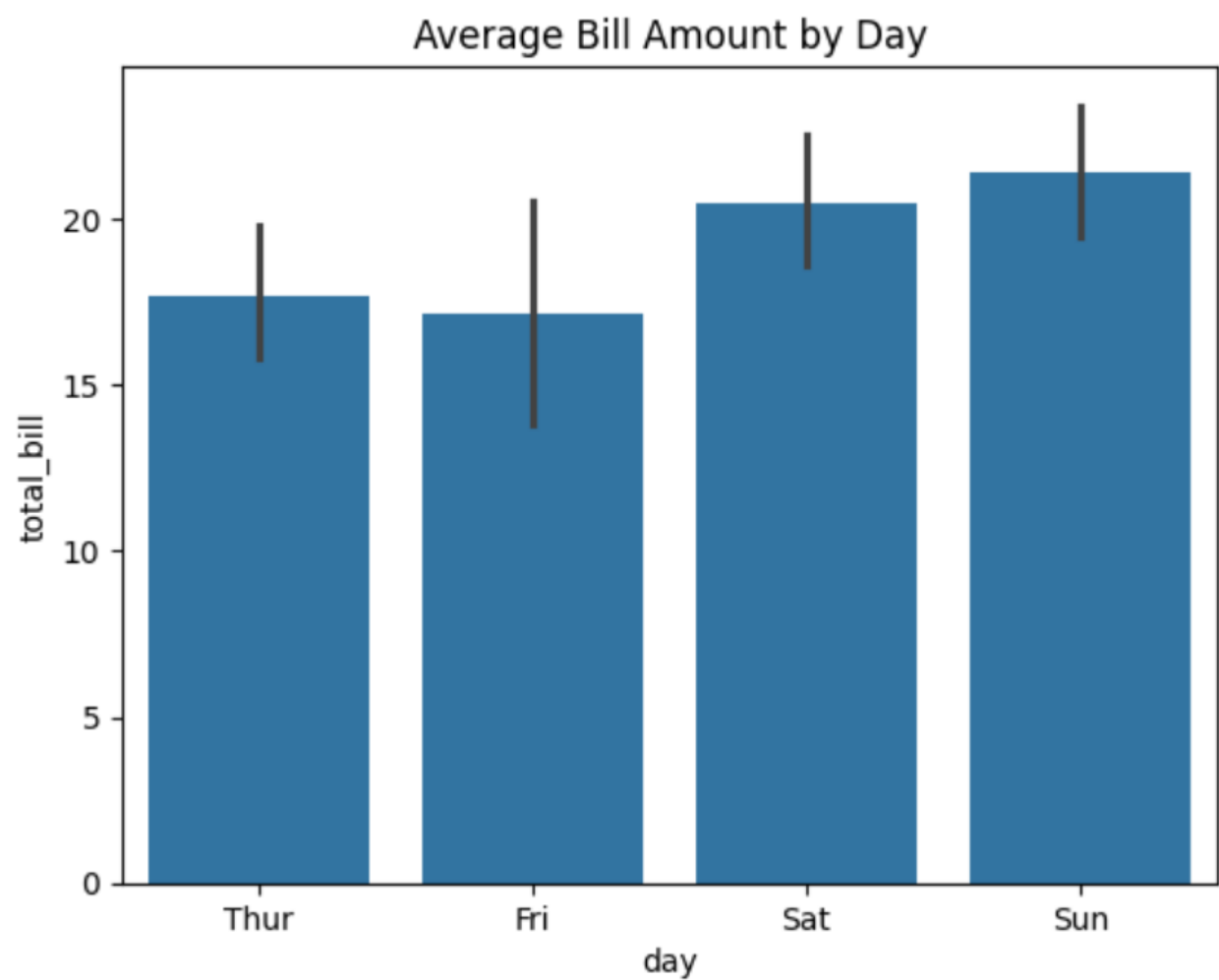
```
tips = sns.load_dataset("tips")
```

```
sns.barplot(x="day", y="total_bill", data=tips)
```

```
plt.title("Average Bill Amount by Day")
```

```
plt.show()
```

Output



HISTPLOT

Description

A histplot is used to visualize the distribution of numerical data by grouping values into bins. It helps understand how data is spread across different ranges.

Use Case

Analyzing age distributions, exam scores, income levels, and customer spending patterns.

Example

```
import seaborn as sns
```

```
import matplotlib.pyplot as plt
```

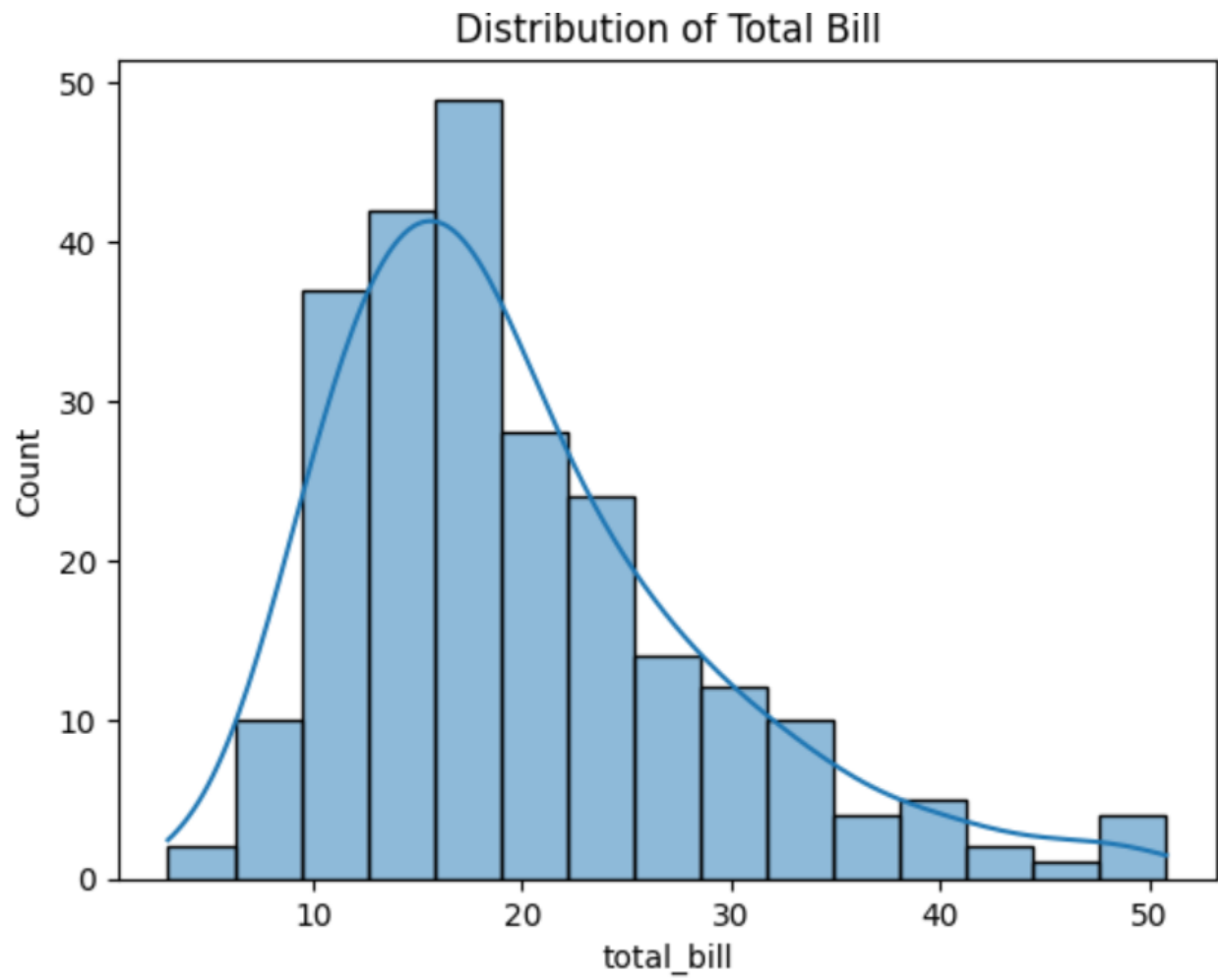
```
tips = sns.load_dataset("tips")
```

```
sns.histplot(tips["total_bill"], bins=15, kde=True)
```

```
plt.title("Distribution of Total Bill")
```

```
plt.show()
```

Output



KDE PLOT

Description

Kernel Density Estimation (KDE) Plot is used to visualize the probability density distribution of continuous data. It provides a smooth curve representing the distribution pattern.

Use Case

Understanding data distribution, identifying peaks, and comparing multiple datasets.

Example

```
import seaborn as sns

import matplotlib.pyplot as plt

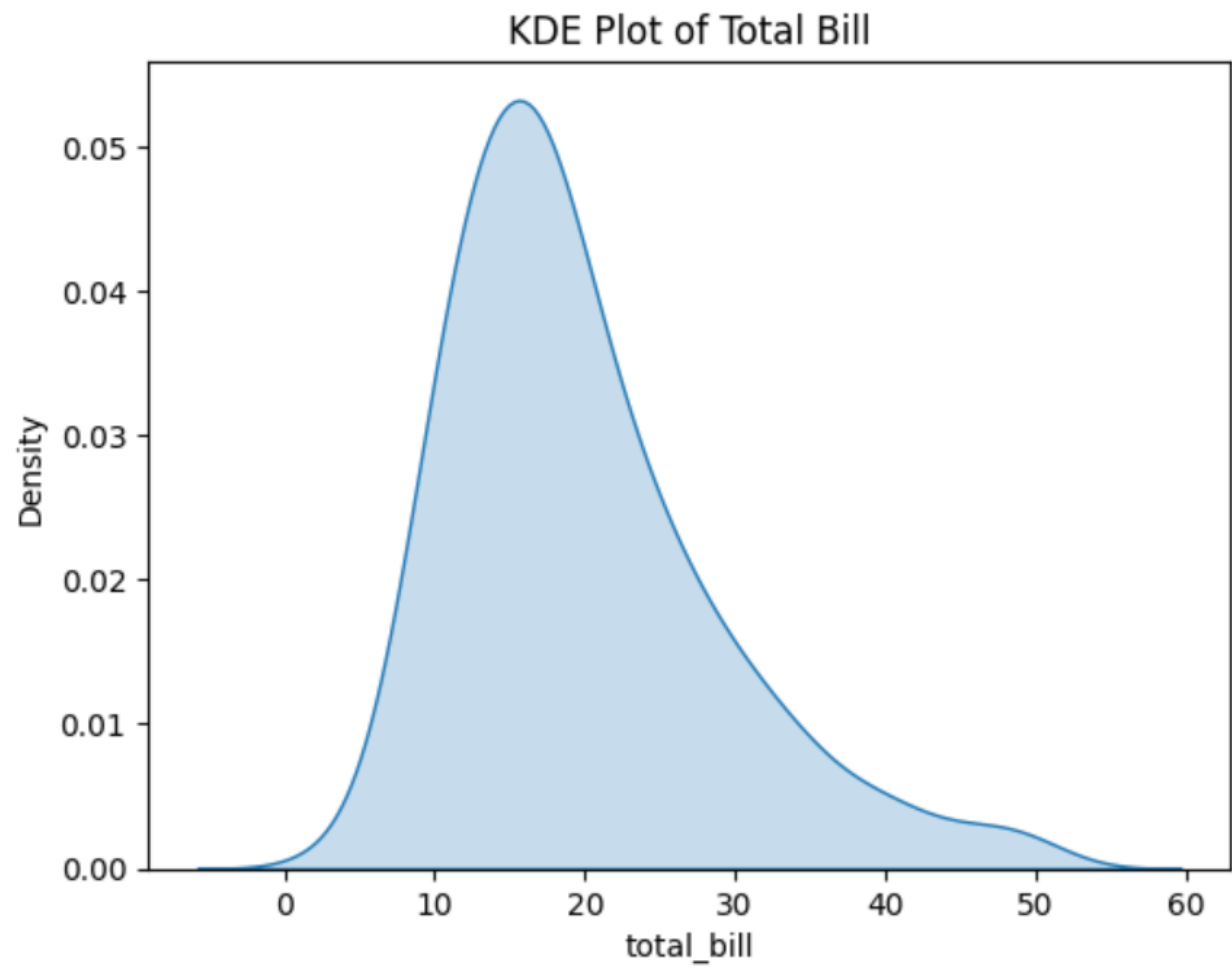
tips = sns.load_dataset("tips")

sns.kdeplot(data=tips, x="total_bill", fill=True)

plt.title("KDE Plot of Total Bill")

plt.show()
```

Output



COMPARISON OF MATPLOTLIB AND SEABORN

Matplotlib

Foundation Library: Matplotlib serves as the foundation for many Python visualization libraries. It provides detailed control over chart elements and supports a broad range of visualization types.

Versatility: The library can be used to create simple graphs as well as advanced visualizations, making it suitable for various analytical tasks.

Customization: Users can modify almost every aspect of a chart, including colors, labels, axes, legends, and annotations.

Advantages of Matplotlib

- Extensive customization capabilities.
- Supports numerous chart types.
- Reliable and widely adopted.
- Suitable for publication-quality visualizations.

Seaborn

Statistical Visualization: Seaborn focuses on creating informative statistical graphics with minimal code.

User-Friendly Design: It offers a simpler interface compared to Matplotlib and provides visually appealing outputs by default.

DataFrame Integration: Seaborn works directly with Pandas DataFrames, simplifying the visualization process.

Advantages of Seaborn

- Easy-to-understand syntax.
- Beautiful built-in themes.
- Strong statistical plotting features.
- Ideal for exploratory data analysis.

Overall Comparison

Matplotlib is an excellent choice when detailed customization and flexibility are required. Seaborn is more suitable for quickly generating attractive and informative statistical visualizations. Both libraries complement each other and are frequently used together in Data Science projects.

CONCLUSION

Matplotlib and Seaborn are among the most widely used visualization libraries in Python. Matplotlib provides flexibility and control, while Seaborn offers simplicity and attractive statistical graphics. Together, these libraries help data professionals explore, analyse, and communicate insights effectively through visual representations.